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CASE STUDY I :-

* Designing a Small Educational RISC-V Based Processor

I) Introduction to the Problem

A startup aims to develop a low cost educational computing kit for engineering students. The objective is to help students understand how instructions are executed inside a processor in a simple and clear way.

The processor must be:

- Easy to understand
- Simple in design
- Cost-effective
- Suitable for learning core CPU concepts.

II) Concepts from COA used

- CPU registers
- Instruction Format
- Instruction Decoding
- Datapath Design
- RISC Architecture

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IV) Proposed Analysis and Design

1) Role of CPU Registers

Registers are small, high-speed storage units inside the CPU used to hold temporary data during execution.

Important Registers:

- Program Counter (PC)
- Instruction Register (IR)
- General Purpose Registers: store operands & results.

2) Instruction format and Decoding

RISC-V uses a fixed 32-bit instruction format, which simplifies hardware design.

Basic Format:-

opcode	rs1	rs2	rd	Add.

Opcode → operation type
rs1, rs2 : source register
rd: destination register
add: data address.

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3) Instruction Cycle

Eg: Instruction: –

ADD R1, R2, R3 → R1 = R2 + R3.

Steps:-

1. Fetch: Instruction is fetched from memory using PC.
2. Decode: Control Unit identifies operation.
3. Execute: ALU performs addition
4. Write Back: Result stored in R1.

4) Addressing Modes and Instruction Types

RISC-V mainly uses simple addressing:

- Immediate Addressing: constant values
- Register Addressing: Operands from registers.
- Load / Store Instruction: memory access

5) Why RISC-V is suitable for education

- Simple and clean instruction set.
- Open source and free to use
- Fixed instruction size → easy decoding
- Less hardware complexity.
- Faster execution due to fewer instructions.

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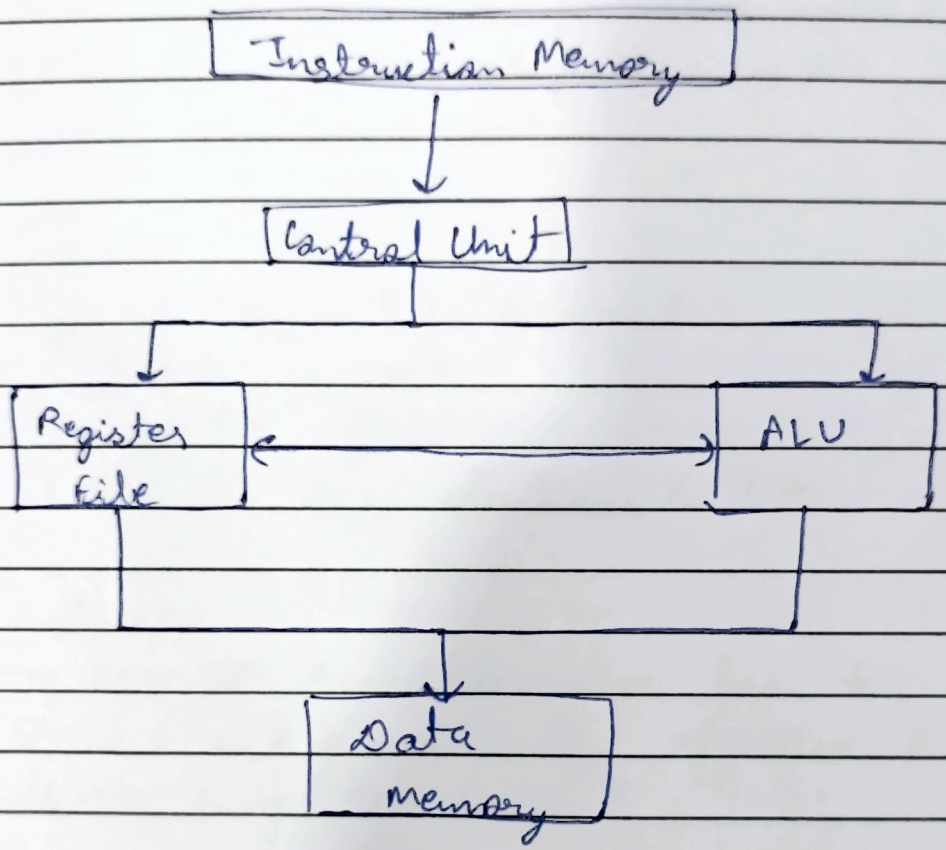
IV) Datapath ~~Dig~~ Diagram
(Single-cycle RISC-V Processor)



This datapath shows how:-

- Instruction are fetched
- Data flows b/w registers & ALU
- Results are stored back.

3) It helps students visually understand processor working.

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V) Final Recommendation

A Single-cycle RISC-V processor with a simple datapath and basic registers is the best choice for an educational kit.

Reasons:-

- Easy to design & understand.
- Low hardware cost.
- Clear execution flow.
- Suitable for demonstrations & labs.

VI) Conclusion

The RISC-V based processor provides a simple, efficient and educationally effective platform for understanding CPU design.

→ It allows students to:

- Visualize instruction execution.
- Learn datapath and control flow
- Build strong fundamentals in computer architecture.

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CASE STUDY II :-

* I/O design for a Hospital Patient Monitoring System

1) Introduction to the Problem

A hospital wants to design a patient monitoring system that continuously tracks:

- Heart rate
- Oxygen level
- Body Temperature.

The system must:

- Process data continuously.
- Generate real-time alerts
- Ensure high reliability and accuracy.

2) Concepts from COA

- Memory-Mapped I/O
- Programmed I/O
- Interrupt-Driven I/O
- Direct Memory Access (DMA)
- Data transfer b/w I/O devices & Memory.
- ~~Memory~~ I/O-Mapped I/O

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3) Proposed Analysis & Design

(a) Memory-Mapped I/O v/s I/O Mapped I/O

Feature	Memory-Mapped I/O	I/O-Mapped I/O
• Address Space	Shared with memory	Separate
• Instructions	Normal memory instructions	Special I/O instructions
• Speed	Faster	Slightly slower
• Complexity	Simple	More complex

(b) Comparison of I/O techniques

1) Programmed I/O
- CPU continuously checks device status
- Data transfer handled by CPU

2) Interrupt-Driven I/O
- Device sends interrupt when ready
- CPU responds only when required

3) DMA
- Data transferred directly b/w device & memory
- CPU involvement is minimal.

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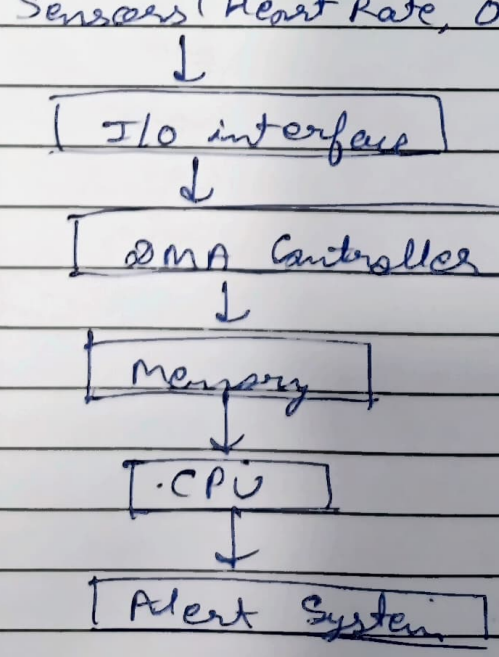
c) Best method Recommendation

→ DMA + Interrupt-Driven I/O [Combined Approach]
- DMA handles continuous bulk data transfer.
- Interrupts handle urgent signals (alerts)

d) Data flow in Patient Monitoring System
Step-by-step flow:-
1) Sensors collect patient data
2) I/O interface converts signals.
3) DMA transfers data to memory
4) CPU processes data.
5) Alert system generates warning if needed.

e) Diagram (I/O sys. with DMA & Interrupts)



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5) Final Recommendation

The best design is:

Memory-Mapped I/O + DMA + Interrupt Sys.

Reasons:-

- Fast data transfer
- Low CPU load
- Real time response
- High reliability.

6) Conclusion

The use of DMA with interrupt driven I/O ensures:

- Continuous monitoring
- Immediate alert generation
- Efficient processor usage

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